Instrument Variables Revisted

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Contents


```
bookdown::render_book()
```

Overview

We will revisit the topic of instrumental variables, but we will take deeper dive into estimates of the $LATE$.

1. Instrumental Variables Revisited
2. Jackknife Instrumental Variables Estimator

Chapter 1

Instrumental Variables

In-depth, institutional knowledge of programs, treatment, interventions provide some of the best sources of strong instruments (Angrist and Krueger, 2001)

We are going to estimate the marginal impact of going to college for those who comply with the instrument.

$$\delta^{IV}_{LATE} = \frac{E[Y_i(D^1_i, 1) - Y_i(D^0_i, 0)]}{E[D^1_i - D^0_i]} = E[Y^1_i - Y^0_i | D^1_i - D^0_i = 1]$$

1.1 Naive OLS

We will use a naive OLS to estimate the marginal effect of college on wages. We expect this to be upwards biased.

First, we will import our data from Card via Cunningham

```
cd "/Users/Sam/Desktop/Econ 672/Data"
use card
```

$$ln(wages_i) = \beta_0 + \beta_1 edu_i + \beta_2 expr_i + \beta_3 Black_i + \beta_4 South_i + \beta_5 married_i + a_i + \varepsilon_i$$

Where a_i are controls for metropolitian statistical areas.

```
reg lwage  educ  exper black south married i.smsa
```

```
      Source |       SS           df       MS      Number of obs   =     3,003
-------------+----------------------------------   F(6, 2996)      =    219.15
       Model |  180.255137         6  30.0425229   Prob > F        =    0.0000
    Residual |  410.705979     2,996  .137084773   R-squared       =    0.3050
-------------+----------------------------------   Adj R-squared   =    0.3036
       Total |  590.961117     3,002  .196855802   Root MSE        =    .37025

------------------------------------------------------------------------------
       lwage |      Coef.   Std. Err.      t    P>|t|     [95% Conf. Interval]
-------------+----------------------------------------------------------------
        educ |   .0711729   .0034824    20.44   0.000     .0643447     .078001
       exper |   .0341518   .0022144    15.42   0.000     .0298098    .0384938
       black |  -.1660274   .0176137    -9.43   0.000    -.2005636   -.1314913
       south |  -.1315518   .0149691    -8.79   0.000    -.1609024   -.1022011
     married |  -.0358707   .0034012   -10.55   0.000    -.0425396   -.0292019
      1.smsa |   .1757871   .0154578    11.37   0.000     .1454782    .2060961
       _cons |   5.063317   .0637402    79.44   0.000     4.938338    5.188296
------------------------------------------------------------------------------
```

We find that our $\hat{\beta}_1 = 0.0711$ and our interpretation is that college increases wages by $(e^{0.0711} - 1) * 100\% = 7.3\%$

1.2 Estimate Local Average Treatment Effect using IV

Next, we will use our instrument variable estimator to estimate the $LATE$ for those who comply with the instrument. We will use the *ivregress 2sls* command.

Our instrument is a binary if being in a county near a 4-year college.

Our first stage is:

$$educ_i = \pi_0 + \pi_1 nearc4_i + \pi_2 expr_i + \pi_3 Black_i + \pi_4 South + \pi_5 married + \pi_6 metro_i + \eta_i$$

Our second state is:

$$ln(wages_i) = \beta_0 + \beta_1 \widehat{edu}_i + \beta_2 expr_i + \beta_3 Black_i + \beta_4 South_i + \beta_5 married_i + \beta_6 metro_i + \varepsilon_i$$

```
ivregress 2sls lwage (educ=nearc4) exper black south married i.smsa, first
```

```
First-stage regressions
-----------------------

                                                Number of obs     =      3,003
```

```
                                                F(   6,   2996)   =     456.14
                                                Prob > F          =     0.0000
                                                R-squared         =     0.4774
                                                Adj R-squared     =     0.4764
                                                Root MSE          =     1.9373

------------------------------------------------------------------------------
        educ |      Coef.   Std. Err.      t    P>|t|     [95% Conf. Interval]
-------------+----------------------------------------------------------------
       exper |   -.404434   .0089402   -45.24   0.000    -.4219636   -.3869044
       black |  -.9475281   .0905256   -10.47   0.000    -1.125027   -.7700295
       south |  -.2973528   .0790643    -3.76   0.000    -.4523787   -.1423269
     married |  -.0726936   .0177473    -4.10   0.000    -.1074918   -.0378954
      1.smsa |   .4208945    .084868     4.96   0.000     .2544891       .5873
      nearc4 |   .3272826   .0824239     3.97   0.000     .1656695    .4888957
       _cons |    16.8307   .1307475   128.73   0.000     16.57433    17.08706
------------------------------------------------------------------------------


Instrumental variables (2SLS) regression          Number of obs   =      3,003
                                                  Wald chi2(6)    =     840.98
                                                  Prob > chi2     =     0.0000
                                                  R-squared       =     0.2513
                                                  Root MSE        =     .38384

------------------------------------------------------------------------------
       lwage |      Coef.   Std. Err.      z    P>|z|     [95% Conf. Interval]
-------------+----------------------------------------------------------------
        educ |   .1241642   .0498975     2.49   0.013     .0263668    .2219616
       exper |   .0555882   .0202624     2.74   0.006     .0158746    .0953019
       black |  -.1156855   .0506823    -2.28   0.022    -.2150211     -.01635
       south |  -.1131647   .0232168    -4.87   0.000    -.1586687   -.0676607
     married |  -.0319754    .005081    -6.29   0.000    -.0419339   -.0220169
      1.smsa |   .1477065   .0308591     4.79   0.000     .0872237    .2081893
       _cons |   4.162476   .8485997     4.91   0.000     2.499251    5.825701
------------------------------------------------------------------------------
Instrumented:  educ
Instruments:   exper black south married 1.smsa nearc4
```

Our 2SLS estimate of the LATE is 13.5%. Being near a college increases the wages by $(e^{0.124} - 1) * 100\% = 13.5\%$ for compliers.

1.3 Test Instrument Relevance

We will first test our instrument relevance assumption. This should be familiar from Econ 645. We will estimate the first stage by itself and then use an F-Test.

$$educ_i = \pi_0 + \pi_1 nearc4_i + \pi_2 expr_i + \pi_3 Black_i + \pi_4 South + \pi_5 married + \pi_6 metro_i + \eta_i$$

```
reg educ nearc4 exper black south married i.smsa
```

```
      Source |       SS           df       MS      Number of obs   =     3,003
-------------+----------------------------------   F(6, 2996)      =    456.14
       Model |  10272.0963         6  1712.01605   Prob > F        =    0.0000
    Residual |  11244.7835     2,996  3.75326552   R-squared       =    0.4774
-------------+----------------------------------   Adj R-squared   =    0.4764
       Total |  21516.8798     3,002  7.16751492   Root MSE        =    1.9373

------------------------------------------------------------------------------
        educ |      Coef.   Std. Err.      t    P>|t|     [95% Conf. Interval]
-------------+----------------------------------------------------------------
      nearc4 |   .3272826   .0824239     3.97   0.000     .1656695    .4888957
       exper |   -.404434   .0089402   -45.24   0.000    -.4219636   -.3869044
       black |  -.9475281   .0905256   -10.47   0.000    -1.125027   -.7700295
       south |  -.2973528   .0790643    -3.76   0.000    -.4523787   -.1423269
     married |  -.0726936   .0177473    -4.10   0.000    -.1074918   -.0378954
      1.smsa |   .4208945    .084868     4.96   0.000     .2544891       .5873
       _cons |    16.8307   .1307475   128.73   0.000     16.57433    17.08706
------------------------------------------------------------------------------
```

Being in a county near a college is associated with a 0.32 increase in years of education.

Next, we will use an F-test on the excludability of college in the county from the first stage regression.

```
test nearc4
```

```
 ( 1)  nearc4 = 0

       F(  1,  2996) =   15.77
            Prob > F =    0.0001
```

The F-statistic is 15.767 which is indicative of a good instrument

1.4 Test Monotonicity

We will now test the monotonicity assumption of the IV. First, we will rerun the first stage regression and predict education.

```
quietly reg educ nearc4 exper black south married smsa
predict dhat
```

Next, we will average outcomes across dhat by using the *egen cut* to get bins of outcomes

```
sum lwage, detail
egen lwage_bins = cut(lwage), at(5,5.25,5.5,5.75,6,6.25,6.5,6.75,7,7.25,7.5)
sort lwage_bins
by lwage_bins: egen mean_z=mean(nearc4)
```

Now we will graph the monotonicity assumption.

```
*Show Monotonicity Graph
sort mean_z
twoway line lwage_bins mean_z
```

It is possible that monotonicity assumption is violated.

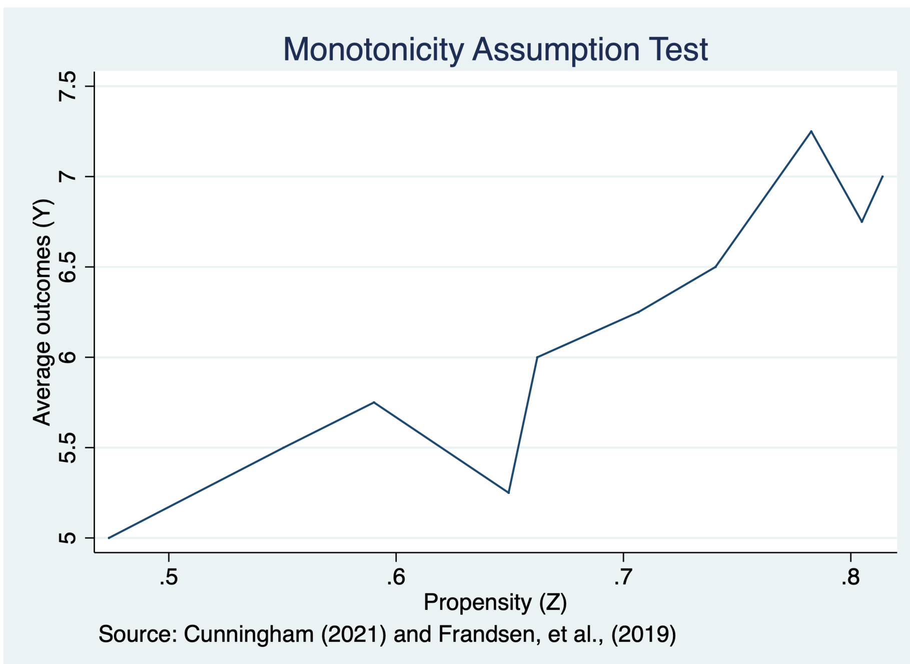

Figure 1.1: Monotonicity Assumption Test

Chapter 2

Jackknife Instrument Variable Estimator

We can use the Jackknife Instrument Variable Estimator (JIVE) for Judget Fixed Effects.

First we will start by installing the Jackknife IV estimator.

```
net install st0108
```

Next we will load our data from Stevenson (2018) on guilt verdicts and cash bail/pretrail detention.

```
cd "/Users/Sam/Desktop/Econ 672/Data"
use judge_fe, clear
```

Next, we will set up some global macros for sets of covariates.

```
*Set up global macros
global judge_pre judge_pre_1 judge_pre_2 judge_pre_3 judge_pre_4 judge_pre_5 judge_pre_6 judge_pr
global demo black age male white
global off      fel mis sum F1 F2 F3 F M1 M2 M3 M
global prior priorCases priorWI5 prior_felChar  prior_guilt onePrior threePriors
global control2     day day2 day3  bailDate t1 t2 t3 t4 t5 t6
```

2.1 Naive OLS

We will estimate a couple specifications using an OLS estimator. Our first model will have minimal controls for time variables, but our second model will have additional controls for demographics and prior justice involvement.

$$guilt_i = \delta_0 + \delta_1 PretrailDentention_i + \gamma' Dates_i + \beta' Demo_i + \alpha' Crime_i + \varepsilon_i$$

```
eststo minimal: quietly reg guilt jail3 $control2, robust
eststo maximum: quietly reg guilt jail3 possess robbery DUI1st drugSell aggAss $demo $p

esttab minimal maximum, mtitle("Minimal" "Max Controls") keep(jail3)
```

```
                      (1)             (2)
                  Minimal    Max Controls
--------------------------------------------
jail3           -0.000735          0.0292***
                  (-0.42)         (15.23)
--------------------------------------------
N                  331971          331971
--------------------------------------------
t statistics in parentheses
* p<0.05, ** p<0.01, *** p<0.001
```

The minimal OLS estimate of pretrail detention on guilty plea is -0.0007 increase in percentage points of a guily plea and not statistically significant at the 5 percent level, while maximum controls yields an estimate of 2.9 percentage points.

2.2 Instrument Variables

We will estimate the reduced form equation for the first stage of the instrument variable estimator. We will estimate the first stage with two reduced form equations. One with date variables t_i and one with data variables plus demographics d_i and crime and/or prior justice involement c_i.

$$PreTrailDetention_i = \pi_0 + \pi_1 JFE_i + t_i + d_i + c_i + \varepsilon_i$$

```
est clear
eststo rf1: quietly reg jail3 $judge_pre $control2, robust
eststo rf2: quietly reg jail3 possess robbery DUI1st drugSell aggAss $demo $prior $off
esttab rf1 rf2, mtitle("Min Controls" "Max Controls") keep(judge_pre_1 judge_pre_2 judg
```

```
                      (1)             (2)
             Min Controls    Max Controls
--------------------------------------------
```

```
judge_pre_1        -0.0325***      -0.0306***
                   (-5.80)         (-6.11)

judge_pre_2              0               0
                       (.)             (.)

judge_pre_3        -0.0237***      -0.0192***
                   (-4.63)         (-4.20)

judge_pre_4        -0.0458***      -0.0442***
                   (-8.97)         (-9.71)

judge_pre_5        -0.0338***      -0.0301***
                   (-5.99)         (-6.02)

judge_pre_6       -0.00910        -0.00740
                   (-1.78)         (-1.62)

judge_pre_7        -0.0307***      -0.0253***
                   (-5.63)         (-5.22)

judge_pre_8        -0.0427***      -0.0361***
                   (-8.36)         (-7.89)
--------------------------------------------
N                   331971          331971
--------------------------------------------
t statistics in parentheses
* p<0.05, ** p<0.01, *** p<0.001
```

Next, we will estimate the F-statistics for our instrument relevance tests for both models.

```
quietly {
reg jail3 $judge_pre $control2, robust
}
test $judge_pre
```

```
 ( 1)  judge_pre_1 = 0
 ( 2)  o.judge_pre_2 = 0
 ( 3)  judge_pre_3 = 0
 ( 4)  judge_pre_4 = 0
 ( 5)  judge_pre_5 = 0
 ( 6)  judge_pre_6 = 0
 ( 7)  judge_pre_7 = 0
 ( 8)  judge_pre_8 = 0
```

```
       Constraint 2 dropped

       F(  7,331954) =   35.25
            Prob > F =    0.0000
```

Our judge fixed effects appear to be relevant instruments since $F - stat = 35.25$ for our minimal controls model.

```
quietly {
reg jail3 $judge_pre possess robbery DUI1st drugSell aggAss $demo $prior $off $control2
}
test $judge_pre
```

```
 ( 1)  judge_pre_1 = 0
 ( 2)  o.judge_pre_2 = 0
 ( 3)  judge_pre_3 = 0
 ( 4)  judge_pre_4 = 0
 ( 5)  judge_pre_5 = 0
 ( 6)  judge_pre_6 = 0
 ( 7)  judge_pre_7 = 0
 ( 8)  judge_pre_8 = 0
       Constraint 2 dropped

       F(  7,331928) =   42.67
            Prob > F =    0.0000
```

Our judge fixed effects appear to be relevant instruments since $F - stat = 42.67$ for our maximum controls model, as well.

Now let's estimate our two models with an IV estimator with the *ivregress 2sls* command.

```
est clear
eststo iv1: quietly ivregress 2sls guilt (jail3= $judge_pre) $control2, robust first
eststo iv2: quietly ivregress 2sls guilt (jail3= $judge_pre) possess robbery DUI1st dru
esttab iv1 iv2, mtitle("Min IV Model" "Max IV Model") keep(jail3)
```

```
                      (1)             (2)
             Min IV Model    Max IV Model
--------------------------------------------
jail3             0.151*          0.186**
                 (2.32)          (2.88)
--------------------------------------------
N                331971          331971
```

```
--------------------------------------------
t statistics in parentheses
* p<0.05, ** p<0.01, *** p<0.001
```

Our *LATE* estimate is an increase of guilty plea by 15.1 percentage points for individuals with a guilty plea with the minimal-controls model.

Our *LATE* estimate is an increase of guilty plea by 18.6 percentage points for individuals with a guilty plea with the maximum-controls model.

2.3 Jackknife Instrument Variable Estimator

Now we will implement our Jackknife IV estimator and estimate miminal-control and maximum-control models. We use the *jive* command after doing our install of package st0108.

```
jive guilt (jail3= $judge_pre) $control2, robust
```

```
note: t6 dropped due to collinearity
note: judge_pre_8 dropped due to collinearity

Jackknife instrumental variables regression (UJIVE1)

First-stage summary                                     Number of obs   =  331971
-------------------------                               F(  10,331960)  =  270.73
F(   7,331954)  =   35.36                               Prob > F        =  0.0000
Prob > F        =  0.0000                               R-squared       = -0.0183
R-squared       =  0.0026                               Adj R-squared   = -0.0183
                                                        Root MSE        =  0.5045

------------------------------------------------------------------------------
       guilt |      Coef.   Std. Err.      t    P>|t|     [95% Conf. Interval]
-------------+----------------------------------------------------------------
       jail3 |   .1624352   .0701409     2.32   0.021      .024961    .2999094
         day |  -.0000271   .0000664    -0.41   0.683    -.0001574    .0001031
        day2 |  -1.18e-07   3.25e-07    -0.36   0.716    -7.56e-07    5.19e-07
        day3 |   2.65e-10   4.23e-10     0.63   0.531    -5.64e-10    1.09e-09
    bailDate |   .0000606   9.00e-06     6.74   0.000      .000043    .0000783
          t1 |   .0094375   .0174909     0.54   0.589    -.0248442    .0437192
          t2 |  -.0118243   .0136833    -0.86   0.388    -.0386431    .0149946
          t3 |   .0101888   .0123284     0.83   0.409    -.0139746    .0343521
          t4 |   .0195249   .0080497     2.43   0.015     .0037478     .035302
          t5 |   .0017162   .0055469     0.31   0.757    -.0091555    .0125879
       _cons |   -.675075   .1748001    -3.86   0.000    -1.017678    -.332472
```

```
------------------------------------------------------------------------------
Instrumented:  jail3
Instruments:   day day2 day3 bailDate t1 t2 t3 t4 t5  judge_pre_1 judge_pre_2
               judge_pre_3 judge_pre_4 judge_pre_5 judge_pre_6 judge_pre_7
------------------------------------------------------------------------------
```

Our estimated $LATE$ with JIVE is a 16.2 percentage point increase in a guilty plea for individuals with a pre-trail detention.

```
jive guilt (jail3= $judge_pre) possess robbery DUI1st drugSell aggAss $demo $prior $of
```

```
note: t6 dropped due to collinearity
note: judge_pre_8 dropped due to collinearity

Jackknife instrumental variables regression (UJIVE1)

First-stage summary                                    Number of obs   =  331971
-------------------------                              F(  36,331934)  = 1375.47
F(   7,331928)  =   42.83                              Prob > F        =  0.0000
Prob > F        =  0.0000                              R-squared       =  0.0925
R-squared       =  0.2486                              Adj R-squared   =  0.0924
                                                       Root MSE        =  0.4763

-------------------------------------------------------------------------------
        guilt |      Coef.   Std. Err.      t    P>|t|     [95% Conf. Interval]
--------------+----------------------------------------------------------------
        jail3 |   .2120002   .0755804     2.80   0.005     .0638648    .3601355
      possess |  -.0609555   .0038514   -15.83   0.000    -.0685041   -.0534068
      robbery |   -.102243   .0091054   -11.23   0.000    -.1200894   -.0843966
       DUI1st |   .0583614   .0066367     8.79   0.000     .0453536    .0713692
     drugSell |   .1154289   .0096137    12.01   0.000     .0965863    .1342715
       aggAss |    .004039   .0040886     0.99   0.323    -.0039746    .0120526
        black |   .0570339   .0061167     9.32   0.000     .0450452    .0690225
          age |    .001521   .0001193    12.75   0.000     .0012871    .0017548
         male |  -.0544413    .008064    -6.75   0.000    -.0702465   -.0386362
        white |   .1002602   .0038903    25.77   0.000     .0926353    .1078851
  priorCases |  -.0057862   .0002551   -22.68   0.000    -.0062861   -.0052862
     priorWI5 |   .0245624   .0060695     4.05   0.000     .0126663    .0364585
prior_felChar |  -.0076116   .0008469    -8.99   0.000    -.0092714   -.0059518
  prior_guilt |    .023201    .001188    19.53   0.000     .0208726    .0255293
     onePrior |   .0473753   .0035872    13.21   0.000     .0403445    .0544062
  threePriors |  -.0015179   .0048536    -0.31   0.754    -.0110307    .0079949
          fel |  -.0220026   .0107113    -2.05   0.040    -.0429965   -.0010087
          mis |   .1393341   .0115767    12.04   0.000     .1166441    .1620242
          sum |   .0657736   .0036656    17.94   0.000     .0585891     .072958
```

```
          F1 |  -.0156777    .012377    -1.27   0.205    -.0399363    .0085808
          F2 |    .027801   .0111437     2.49   0.013     .0059597    .0496423
          F3 |   .0923061   .0031874    28.96   0.000      .086059    .0985532
           F |  -.0929481   .0119417    -7.78   0.000    -.1163534   -.0695427
          M1 |   .0094806   .0076985     1.23   0.218    -.0056083    .0245695
          M2 |  -.0772227   .0045121   -17.11   0.000    -.0860662   -.0683791
          M3 |   .1242172   .0051198    24.26   0.000     .1141825    .1342518
           M |   .2706689   .0053935    50.18   0.000     .2600978      .28124
         day |    -.00003   .0000608    -0.49   0.622    -.0001492    .0000892
        day2 |  -7.03e-09   2.95e-07    -0.02   0.981    -5.85e-07    5.71e-07
        day3 |   2.42e-10   3.87e-10     0.63   0.532    -5.16e-10    9.99e-10
    bailDate |   .0000418   8.57e-06     4.88   0.000      .000025    .0000586
          t1 |  -.0186899    .016402    -1.14   0.255    -.0508374    .0134576
          t2 |  -.0243829   .0127911    -1.91   0.057     -.049453    .0006872
          t3 |   .0033173   .0110964     0.30   0.765    -.0184314     .025066
          t4 |   .0142448   .0072415     1.97   0.049     .0000517     .028438
          t5 |    .009051   .0052937     1.71   0.087    -.0013245    .0194264
       _cons |  -.6960786    .164429    -4.23   0.000    -1.018355   -.3738024
------------------------------------------------------------------------------
Instrumented:  jail3
Instruments:   possess robbery DUI1st drugSell aggAss black age male white
               priorCases priorWI5 prior_felChar prior_guilt onePrior
               threePriors fel mis sum F1 F2 F3 F mis M1 M2 M3 M day day2 day3
               bailDate t1 t2 t3 t4 t5  judge_pre_1 judge_pre_2 judge_pre_3
               judge_pre_4 judge_pre_5 judge_pre_6 judge_pre_7
------------------------------------------------------------------------------
```

Our estimated $LATE$ with JIVE is a 21.2 percentage point increase in a guilty plea for individuals with a pre-trail detention

Also, don't forget to clear your global macros! Local macros expire at the end of a run, but global macros will remain, and you need to tell Stata to remove them.

```
clear all
macro drop _all
```

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Stevenson, M.T. (2018). Distortion of justice: How the inability to pay bail affects case outcomes. *Journal of Law, Economics, and Organization.* 34(4) p.511-542.